

Diffusive and Bending Modes of the S&P 500 Implied-Volatility Surface: A Field-Theory Factor Model Tested Out of Sample

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github.com/MichaelAafton/diffusion-modes-vol-surface

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Abstract

We test whether the day-to-day dynamics of the S&P 500 implied-volatility smile are consistent with a stochastic heat equation, using ten years (2015–2024) of OptionMetrics IvyDB volatility surfaces. A validated synthetic harness establishes the correct null prediction: for daily *changes*, an Ornstein–Uhlenbeck mode obeys $\text{Var}(\Delta f_k) = 2V_k(1 - e^{-\gamma_k dt})$, not the naïve k^{-2} stationary law. Against this corrected null, pure diffusion is decisively rejected: across a matched-instrument parameter sweep, its spectral slope saturates at $p \leq 1.87$, versus $p \approx 2.5$ – 2.6 observed in the tail of the real spectrum. A three-parameter composite model—an external volatility-level factor riding on a membrane with both tension and bending rigidity, $\gamma_k = Dk^2 + \kappa k^4$ —reproduces the full eight-point empirical spectrum. Formal calibration with block-bootstrap errors yields $D = 0.69$ [0.29, 1.25], $\kappa = 0.011$ [0.006, 0.020], and a diffusion-to-bending crossover $k^* = 7.9$ [5.1, 11.2] at the upper edge of the observable band ($\chi^2/\text{dof} = 3.6$). The sole significant residual is a mode-5 “shelf” ($z = -3.0$). In a chronological out-of-sample test (train through 2022-01-05, test through 2024-12-31), the three-parameter field theory predicts portfolio risk within 1% of an empirical three-factor model on level, straddle, and risk-reversal exposures, while a butterfly portfolio—loading precisely modes 3–5—exposes the same spectral residual in money terms, over-predicted $2.2\times$. The smile behaves as a stiff membrane plus an external level factor; where the picture fails is localised, twice measured, and characterised.

1 Introduction

The implied-volatility surface of a liquid index [3] is not static: it breathes daily, and the structure of that breathing determines the risk of every option portfolio. A long-standing physics-inspired proposal [4, 1, 2]¹ treats the surface as an elastic membrane: local supply–demand shocks perturb it, and an effective stiffness propagates and damps the perturbations. In its simplest form the proposal is a stochastic heat equation,

$$\partial_t p(z, t) = D \partial_z^2 p(z, t) + \xi(z, t), \quad (1)$$

for the perturbation field p on a moneyness axis z , with white noise ξ . Decomposed in Fourier modes, each amplitude is an independent Ornstein–Uhlenbeck (OU) process with relaxation rate $\gamma_k = Dk^2$, so the stationary mode variances scale as k^{-2} and principal-component analysis (PCA) of surface co-movement should recover sinusoidal eigenvectors with a k^{-2} eigenvalue decay.

¹The project was directly inspired by an unpublished presentation, *A Data-Driven Factor Model for Option Risk*, by P. Ioselevich (Capital Fund Management), for which no citable published version exists.

This paper tests that prediction against real S&P 500 (SPX) surfaces—and reports, as its central results, the two places the prediction fails and the minimal extension that repairs them. The design principle throughout is that synthetic data is a *validation harness*, never evidence: simulations prove the pipeline recovers known ground truth; only real data can support or falsify the physical hypothesis; and deviations from the diffusive picture are the scientific product, not a defect.

The contributions are:

1. **A corrected null hypothesis.** The k^{-2} law governs mode *levels*; PCA is necessarily run on daily *changes* (for stationarity), whose spectrum obeys the OU increment law $\text{Var}(\Delta f_k) = 2V_k(1 - e^{-\gamma_k dt})$ —flat for slow modes, $\propto k^{-2}$ for fast ones. Testing real data against the wrong null materially changes conclusions (Sec. 4).
2. **Falsification of pure diffusion.** Through a matched instrument (identical grid, horizon, differencing, and estimator for synthetic and real data), no diffusion constant reproduces the observed spectrum: full-spectrum slope saturates at $p \leq 1.78$ and tail slope at $p \leq 1.87$, versus real values of 2.99–3.20 and 2.53–2.64 (Sec. 6).
3. **A three-parameter composite that works.** An external level factor plus a membrane with bending rigidity, $\gamma_k = Dk^2 + \kappa k^4$, reproduces the eight-point spectrum, including an untuned match of the smallest eigenvalue. Formal calibration with block-bootstrap uncertainties places the diffusion-to-bending crossover at the top of the observable band (Sec. 7).
4. **An out-of-sample risk test through a regime change.** Trained before 2022 and tested on 2022–2024, the three-parameter theory matches a 36-parameter empirical factor model to within 1% on three of four standard portfolios; the fourth (butterfly) independently rediscovers the calibration’s one significant residual in money terms (Sec. 8).

Section 2 develops the theory, Sec. 3 the data and instrument, and Sec. 9 discusses limitations, the decision log, and future work.

2 Theoretical framework

2.1 Modes, levels, and the increment law

On a finite z -domain of length L with the operator $\mathcal{L}$, the field decomposes onto the cosine basis $\varphi_k(z) = \sqrt{2/L} \cos(k\pi(z - z_{\min})/L)$, $k = 1, \dots$. Each amplitude obeys an OU equation $\dot{f}_k = -\gamma_k f_k + \xi_k$, whose stationary variance is

$$V_k = \frac{\sigma^2}{2\gamma_k}. \quad (2)$$

For pure diffusion $\gamma_k = Dk^2$, giving the k^{-2} level spectrum. We consider the extended operator

$$\gamma_k = Dk^2 + \kappa k^4, \quad (3)$$

in which κ is a bending rigidity (the operator of a stiff plate rather than a tense membrane). The two regimes exchange dominance at the crossover wavenumber

$$k^* = \sqrt{D/\kappa}. \quad (4)$$

Volatility levels are nearly integrated (close to a random walk), so PCA is performed on daily changes $\Delta\sigma$, which are approximately stationary. The correct null for a change-based spectrum

follows from the exact OU transition $f(t + dt) = f e^{-\gamma dt} + \eta$, $\eta \sim \mathcal{N}(0, V(1 - e^{-2\gamma dt}))$: writing $a = e^{-\gamma dt}$ and using independence of η from f ,

$$\text{Var}(\Delta f) = V(1 - a)^2 + V(1 - a^2) = 2V(1 - e^{-\gamma dt}). \quad (5)$$

Equation (5) has two personalities: for slow modes ($\gamma dt \ll 1$) it reduces to $\sigma^2 dt$, *independent of k* —all slow modes take the same daily kick, flattening the spectrum’s head—while fast modes ($\gamma dt \gg 1$) decorrelate fully between observations and inherit the level law $2V_k \propto \gamma_k^{-1}$. Any inference that compares a change-based empirical spectrum with the level law conflates these regimes; correcting this changed our own conclusions materially (Sec. 6).

2.2 Composite model

Empirically (Sec. 5) one mode dominates changes with a near-uniform profile across z : the entire smile repricing with the volatility level. We model it as an *external* scalar factor $f_0(t)$ —an AR(1) with its own persistence and daily-kick variance m , added uniformly across z —rather than as the operator’s $k=0$ mode. The model covariance of daily changes at grid points z_i, z_j is then an exact finite sum,

$$\text{Cov}(\Delta p_i, \Delta p_j) = \sum_{k=1}^n \varphi_k(z_i) \varphi_k(z_j) 2V_k(1 - e^{-\gamma_k dt}) + m, \quad (6)$$

which, correlation-normalised and eigendecomposed, yields deterministic model variance shares in microseconds—the forward model used for calibration, with no simulation in the optimisation loop.

3 Data and instrument

3.1 Source and design decisions

Data are OptionMetrics IvyDB smoothed volatility surfaces (WRDS, `vsurf`) for SPX (secid 108105), 2015–2024 ($\sim 2,500$ trading days), pulled once and frozen to Parquet; spot from `secprd`. The surface supplies interpolated implied volatilities on fixed (maturity, delta) pillars together with the vendor’s implied strike. We study the 30-calendar-day maturity slice. Key measurement decisions (the project maintains a numbered decision log, D1–D9; Appendix A):

- **Strike from vendor impl_strike** (D1): self-consistent with the vendor’s own surface construction; fewest moving assumptions.
- **OTM-only stitching** (D2): retain $|\Delta| \leq 0.5$, so out-of-the-money puts populate $z < 0$ and OTM calls $z > 0$ —each half of the axis represented by its liquid instrument.
- **Calendar→business-day conversion** (D3): surface maturities are calendar days; the pipeline’s annualisation uses 252. Omitting the 252/365 conversion silently shrinks every z by $\sim 17\%$.
- **ATM-normalised moneyness** (D4): $z = \log(K/S)/(\sigma_{\text{ATM}}\sqrt{T})$ with σ_{ATM} the per-(date, maturity) 50-delta call/put average. Normalising by each option’s *own* implied volatility makes the coordinate self-referential (a relabelling of delta); the ATM ruler is essential. The missing- σ_{ATM} path raises a hard error rather than falling back silently.

The delta-pillar grid ($|\Delta| \in [0.10, 0.90]$, steps of 0.05) maps to a usable window $z \in [-1.15, 1.10]$; 13.3% of raw rows (nearly all deep OTM puts) lie outside it. The pillar spacing supports ≈ 8 uniform bins—finer binning would manufacture resolution the instrument does not possess. *The instrument’s field of view and mode ceiling are properties of the data source, not of the market*; modes beyond $k \approx 8$ are unobservable here.

Table 1: Empirical variance shares of daily $\Delta\sigma$ across 8 z -bins (2,498 changes), with moving-block bootstrap uncertainty (block = 25 days, 1,000 replicates).

Mode	Share	Boot. std	95% CI
1	0.9150	0.0062	[0.9043, 0.9286]
2	0.0493	0.0056	[0.0358, 0.0552]
3	0.0164	0.0041	[0.0109, 0.0260]
4	0.0067	0.0013	[0.0058, 0.0108]
5	0.0064	0.0007	[0.0046, 0.0071]
6	0.0029	0.0005	[0.0020, 0.0041]
7	0.0019	0.0003	[0.0014, 0.0024]
8	0.0014	0.0002	[0.0010, 0.0018]

3.2 Panel construction

Each option’s z is binned (8 uniform bins on $[-1.15, 1.10]$) and implied vols averaged per (date, bin). A pre-registered 95% coverage rule removes underpopulated bins; with clean numeric dtypes the panel is complete: $2,499 \times 8$ with no missing cells. (An earlier “seven-bin” panel was traced to vendor `Decimal/None` semantics silently corrupting coverage counts; column types are now coerced and unit-tested at the loader.) The analysis variable is the daily change $\Delta\sigma(z)$ (D5): stationary, matched to the harness-validated increment theory, and free of a hedging-frequency assumption. The caveat is inherited honestly: daily $\Delta\sigma$ at fixed z differs from delta-hedged P&L by gamma–theta carry, approximately a rank-one contamination concentrated in the level mode; interpretation of mode 1 carries an asterisk, modes ≥ 2 are robust.

4 Validation harness

The simulator evolves each mode by the *exact* OU transition (zero discretisation error; unconditionally stable; initialised in stationarity), reconstructing $p(z, t)$ on the sampled cosine basis, with the level factor gated behind a configuration flag. The harness’s purpose is falsification-grade validation of the pipeline: simulate, difference, run the *identical* PCA, and require the empirical spectrum to match Eq. (5)’s prediction—which it does to $\lesssim 0.2$ percentage points per mode over 20,000 synthetic days; the comparison is frozen as a permanent unit test. Two methodological rules emerged and are enforced throughout: (i) *aliasing control*: the number of simulated modes equals the number of grid points ($n_{\text{modes}} = n_z$), since coarse sampling folds unrepresented high- k variance into low modes and spuriously steepens spectra; (ii) numerical guards are derived, not inherited—exponential underflow ($e^{-\gamma dt} \rightarrow 0$) is the *benign* exact-update limit (the mode redraws from its stationary law), unlike the instability of Euler–Maruyama at large γdt .

5 The empirical spectrum

Table 1 reports the correlation-matrix PCA of the real $\Delta\sigma$ panel. Structural findings:

Mode shapes. Mode 1 is uniform and positive across z (entries $0.342\text{--}0.366 \approx 1/\sqrt{8}$; zero sign changes): the level mode, owning 91.5% of daily-change variance. Mode 2 is a one-node see-saw, put wing against call wing: the skew mode. Both are stable under removal of the outermost put bin. Mode 3 fails the harmonic test: its shape is a dipole concentrated on the outermost bins that *follows the boundary* when the edge bin is removed, and its node count is unstable ($2 \rightarrow 4$). Physics lives at a location; artifacts live at the boundary. Node-counting evidence for membrane structure therefore stops, by diagnosis rather than oversight, at mode 2. Both diagnostics are shown in Fig. 1.

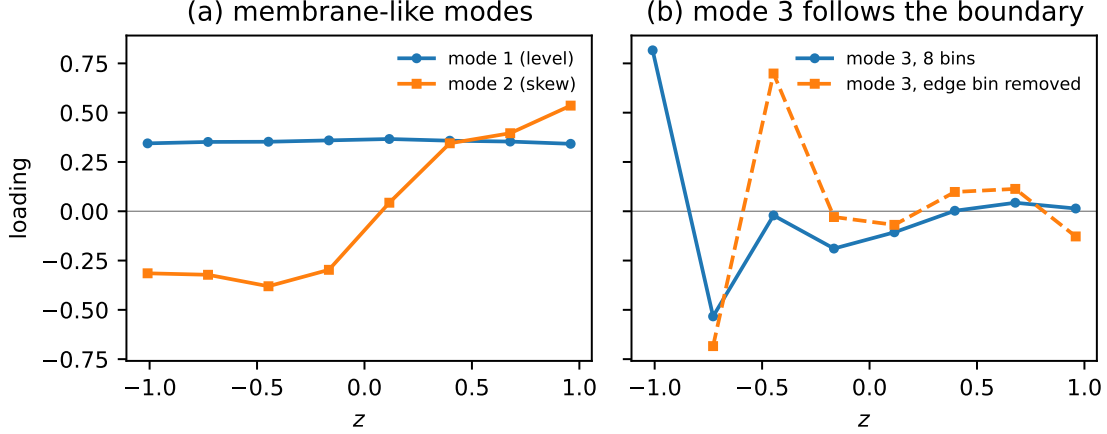


Figure 1: (a) Modes 1–2 of the real $\Delta\sigma$ panel: uniform level and one-node skew see-saw. (b) Mode 3 in the 8-bin panel and with the edge bin removed: the dipole tracks the boundary, identifying it as an edge artifact.

Table 2: Matched-instrument sweep: fitted slopes of synthetic spectra through the identical pipeline (8 bins on $[-1.15, 1.10]$, 2,500 days, 5 seeds per cell, mean $\pm$ s.d.). Top: pure diffusion. Bottom: fixed $D = 0.2$, increasing bending rigidity κ .

D	κ	p_{full}	p_{tail}	k^*
0.01	0	0.36 ± 0.02	0.42 ± 0.03	∞
0.05	0	0.26 ± 0.01	0.34 ± 0.02	∞
0.2	0	0.89 ± 0.02	1.12 ± 0.04	∞
1	0	1.55 ± 0.02	1.73 ± 0.04	∞
5	0	1.78 ± 0.03	1.87 ± 0.04	∞
0.2	0.01	1.54 ± 0.02	2.09 ± 0.04	4.47
0.2	0.03	2.04 ± 0.02	2.74 ± 0.04	2.58
0.2	0.1	2.64 ± 0.02	3.39 ± 0.04	1.41
0.2	0.2	2.97 ± 0.02	3.69 ± 0.04	1.00
0.2	0.3	3.15 ± 0.02	3.83 ± 0.04	0.82
0.2	0.5	3.36 ± 0.02	4.00 ± 0.03	0.63
0.2	1	3.62 ± 0.02	4.19 ± 0.03	0.45

Slopes. A log-log power-law fit $\lambda_k \sim k^{-p}$ over all modes gives $p = 2.99$ (8 bins) and 3.20 (edge bin excluded). Restricted to the tail (modes 2–8, fitted against their true wavenumbers), $p_{\text{tail}} = 2.53$ and 2.64 respectively. The full-spectrum slope is substantially manufactured by mode 1’s dominance; the tail slope is the operator-sensitive statistic. Both exceed the diffusive prediction—but by how much requires the matched-instrument null of the next section, since Eq. (5) distorts any slope measured on daily changes. Fig. 2 shows the data, the calibrated fit of Sec. 7, and the falsified diffusive null in one frame.

6 Operator adjudication

Table 2 answers the confound directly: what slope *would* each candidate operator produce through this exact instrument?

Diffusion is falsified twice. Over the full D sweep, the pure-diffusion slope saturates at $p_{\text{full}} \leq 1.78$ and $p_{\text{tail}} \leq 1.87$: the increment-law flattening can only pull slopes *below* the level-law

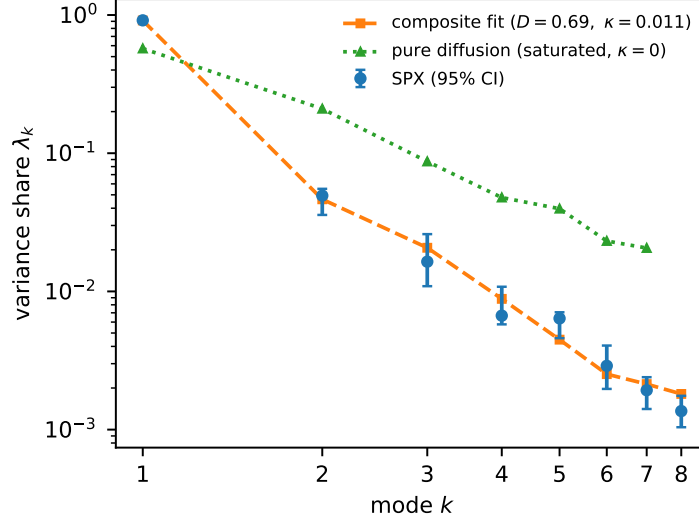


Figure 2: Variance shares of daily $\Delta\sigma$: SPX with 95% block-bootstrap CIs, the calibrated composite model, and pure diffusion at its saturating best (modes 1–7; its eighth eigenvalue is a structural zero of the 8-point sampling). Mode 5 sits above the fit — the shelf residual of Sec. 7.

value of 2, never above. The real values (2.99–3.20; tail 2.53–2.64) are unreachable—a gap of tens of standard errors—whether or not the suspect level mode is included.

A single homogeneous k^4 operator is slope-sufficient but profile-falsified. Bending rigidity does steepen spectra through the observable band, and $\kappa \approx 0.2$ – 0.3 matches the real full-spectrum slope. But the match is an artifact of slope’s shape-blindness: every homogeneous-operator spectrum in the sweep is *concave* on log–log axes ($p_{\text{tail}} > p_{\text{full}}$), while the real spectrum is *convex* ($p_{\text{tail}} < p_{\text{full}}$): a giant head over a gentle tail. At matched slope, the single-operator model mispredicts the mode-2 share sixfold (0.28 vs 0.049). Convexity of the real spectrum forces a two-component structure.

The composite reproduces the eight-point profile. With the level factor tuned to mode 1’s share (0.9157 at daily kick s.d. 3.35, hand stage) and the tail matched by $\kappa \approx 0.02$ at $D = 0.2$, the model tracks the real spectrum mode by mode, including two unforced agreements: the smallest eigenvalue—structurally zero for an 8-mode field on 8 points (rank deficiency), filled to 0.0013 vs real 0.0014 by the level factor as ninth variance source—and near-matching mid-tail shares. Residuals against synthetic seed noise alone appeared enormous (mode 3: $+31\sigma$) but shrink to honest size under the real spectrum’s block-bootstrap errors (mode 3: $+1.9\sigma$, not significant), a caution recorded in the decision log: *seed spread is not measurement error*.

7 Calibration

Parameters are fitted by weighted least squares of the analytic forward model, Eq. (6), against the real shares: $\chi^2 = \sum_{k=1}^7 [(\hat{\lambda}_k - \lambda_k)/\sigma_k]^2$ with block-bootstrap σ_k ; mode 8 is excluded as sum-determined (7 points, 3 parameters, dof = 4). Optimisation is Nelder–Mead in $(\log D, \log \kappa, \log m)$ —positivity by construction, scale-symmetric uncertainty. Confidence intervals come from refitting each of the 1,000 bootstrap spectra.

Table 3 gives the result. Three readings:

The hand-fit anchor was wrong, and the fit says so. The exploratory stage had fixed $D = 0.2$ by inheritance from the sweep and tuned only κ ; freeing all three parameters improves χ^2 by 17.1 (decisive for 3 parameters), trading $3.5\times$ more diffusion against half the bending. $\log D$ and $\log \kappa$ are only weakly correlated across bootstrap refits ($\rho = +0.32$): the data identify

Table 3: Calibrated composite model (bootstrap medians; 95% CIs from bootstrap refits).

Quantity	Estimate	95% CI
D (tension/diffusion)	0.69	[0.29, 1.25]
κ (bending rigidity)	0.011	[0.006, 0.020]
level daily kick $\sqrt{m}$	2.64	[2.25, 3.35]
crossover $k^* = \sqrt{D/\kappa}$	7.9	[5.1, 11.2]
χ^2/dof	3.6	—

Table 4: Out-of-sample variance ratios (realised/predicted; 1 = perfect). Realised variances in units of 10^{-4} .

Portfolio	Realised	Emp. 1f	Emp. 3f	Emp. 5f	Field theory / Diagonal
Straddle	0.62	0.459	0.462	0.463	0.461 / 0.905
Risk reversal	0.12	0.794	0.649	0.649	0.642 / 0.176
Butterfly	0.10	0.906	1.101	1.162	0.495 / 0.072
Equal weight	0.53	0.435	0.438	0.439	0.441 / 3.149

the parameters separately; the intervals are honestly wide rather than degenerate.

The crossover sits at the top of the observable band. $k^* = 7.9$ [5.1, 11.2]: the observable modes are diffusion-dominated, with bending switching on in the last resolvable mode or two. Whether bending fully develops beyond $k \approx 8$ is outside this instrument’s field of view—the resolution ceiling of Sec. 3 becoming a physics limit.

One residual survives. With bootstrap-weighted errors, the fitted model matches all shares except mode 5: real 0.0064, model 0.0044, $z = -3.0$. The real spectrum has a near-degenerate “shelf” at modes 4–5 (0.0067, 0.0064) that no smooth level-plus-operator composite reproduces; it is the leading unexplained feature. ($\chi^2/\text{dof} = 3.6$ quantifies the same statement globally. The unfitted mode-8 share sits at $z = +2.3$, a consistency indicator only.)

8 Out-of-sample risk test

The practical claim of a factor model is predictive: portfolio variance $w^\top \hat{\Sigma} w$ estimated before the fact. We split chronologically—train 2015-01-05 to 2022-01-05 (1,748 changes, COVID-2020 included), test 2022-01-06 to 2024-12-31 (750 changes, an unseen bear-market regime)—and compare three covariance models built from training data only: (i) an empirical PCA factor model (leading factors plus diagonal residual top-up, preserving per-bin variances); (ii) the field theory: model correlations from the calibrated Eq. (6), rescaled by training per-bin standard deviations; (iii) a diagonal (independent-bins) baseline. All three share the training standard deviations, so the contest isolates *correlation structure*—precisely what the theory claims to know from three parameters. Test portfolios are transparent vega-sketch weight vectors on the 8 bins: equal-weight (level), ATM straddle, risk reversal (long put wing/short call wing), and butterfly (wings minus body). The reported statistic is realised test variance over predicted variance (unity is perfect).

Table 4 contains four findings:

A common scale factor, explained. All structured models run at ratios 0.44–0.65: they over-predict, because the training window contains the most violent volatility-of-volatility episode of the decade (March 2020) while the test regime is calmer. The mis-scaling is shared by construction (common training variances) and is a statement about non-stationarity of overall scale, not about correlation structure; dynamic variance rescaling is standard practice and listed as future work.

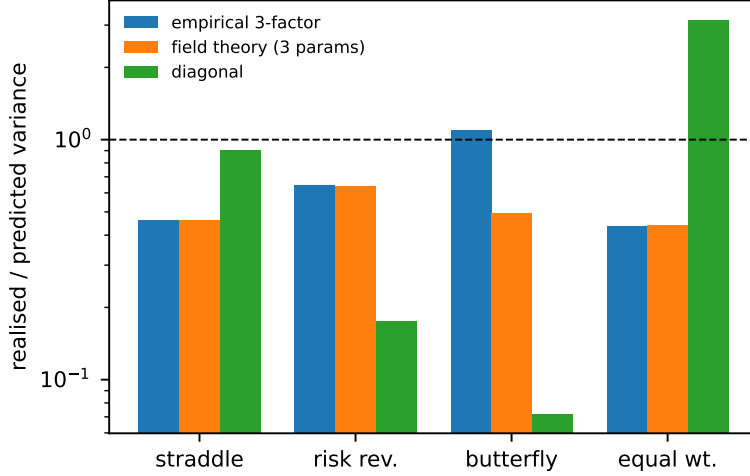


Figure 3: Out-of-sample variance ratios (log scale; dashed line = perfect). Field theory and the empirical 3-factor model agree to $\leq 1\%$ everywhere except the butterfly, which isolates the mode-4–5 residual.

Correlation structure is the whole game. The diagonal baseline fails in *both directions at once*: it under-predicts level risk $3.1\times$ (summing portfolios collect the correlation mass it discards) and over-predicts spread risk $6\text{--}14\times$ (long–short portfolios cancel correlations it never removes).

Three parameters match thirty-six numbers. On level, straddle, and risk-reversal exposures the field theory agrees with the empirical three-factor model to within 1% (e.g. 0.461 vs 0.462), robustly across empirical factor counts (3 vs 5 identical to three decimals; the 1-factor model fails the risk reversal, confirming mode 2 carries real skew risk). Through a regime change, the calibrated physics reproduces the empirical covariance’s out-of-sample predictions on every portfolio that does not isolate the known residual.

The butterfly finds the shelf, independently. The butterfly’s weights annihilate modes 1 (sum zero) and 2 (symmetry), loading precisely modes 3–5—exactly where calibration located its only significant residual. There the field theory over-predicts $2.2\times$ (0.495 vs empirical 1.101). Two measurements of entirely different type—a spectral fit and an out-of-sample money-terms risk ratio—point at the same defect, establishing the mode-4–5 shelf as a property of the surface rather than of any fitting procedure. Fig. 3 summarises the test.

9 Discussion

What holds. Within its observable band, the SPX smile’s daily dynamics are quantitatively consistent with a two-component structure: an external volatility-level factor (91.5% of change variance, uniform in z , plausibly containing the gamma–theta carry and market-wide repricing) riding on a stiff membrane whose relaxation spectrum $\gamma_k = Dk^2 + \kappa k^4$ has its diffusion-to-bending crossover at the edge of resolution. The structure is not merely descriptive: three calibrated parameters predict out-of-sample portfolio risk as well as a full empirical factor model wherever the model’s one known residual is not isolated.

What breaks, and where. Pure diffusion is excluded outright. The homogeneous single-operator picture is excluded by spectral convexity. The composite’s failure is localised to a mode-4–5 shelf, measured twice (spectral $z = -3.0$; butterfly $2.2\times$). Candidate mechanisms—a pinned or resonant structure in the put wing, fat-tailed shocks, slow parameter drift—are deliberately left for future work rather than fitted post hoc.

Limitations. (i) Resolution: 8 bins on $z \in [-1.15, 1.10]$; deep wings and $k > 8$ are invisible; claims are conditional on the band. (ii) Sufficiency, not uniqueness: other mechanisms could

reproduce a finite-resolution spectrum; the claim is that the composite is the minimal sufficient extension tested. (iii) The measurement variable is $\Delta\sigma$, not delta-hedged P&L; the difference is approximately rank-one and confined to mode 1’s interpretation. (iv) Overall variance scale is non-stationary (COVID-in-train); the OOS test controls for it by design but a deployable risk model would rescale dynamically. (v) Maturity structure: this is a fixed-maturity (30-day) slice; the τ -dimension and full 2D membrane are deferred. (vi) Rolling-window stability of the modes (in-sample stability is necessary but not sufficient) is the principal deferred robustness check.

Methodological stance. Every stage was validated against known ground truth before use (pipeline on synthetic data; forward model against simulation; error bars by block bootstrap respecting volatility clustering); predictions were stated before runs and graded after, including the supervisor-side failures (a predicted curvature mode 3, killed by an edge-tracking diagnostic; a predicted strong D - κ degeneracy, not observed); and all measurement choices are recorded in a numbered decision log (Appendix A). Where the data contradicted expectation—the increment-law correction, the seven-bin panel artifact, the mode-3 boundary dipole, the hand-fit’s rejection—the contradiction is reported as part of the result.

A Decision log (abridged)

- D1** Strike from vendor `impl_strike` (self-consistency; fewest assumptions).
- D2** OTM-only put/call stitching, $|\Delta| \leq 0.5$.
- D3** Calendar→business days ($\times 252/365$); omission shrinks z by $\sim 17\%$.
- D4** ATM-normalised z ; usable window $[-1.15, 1.10]$, 8 bins; 13.3% of rows outside; hard error if σ_{ATM} absent.
- D5** Panel variable: daily $\Delta\sigma$ (stationarity; harness-matched; no hedging-frequency assumption); rank-one gamma–theta caveat on mode 1.
- D6** Phase-3 results: level and skew modes robust; mode 3 an edge artifact (node count unstable $2 \rightarrow 4$ under bin removal); slopes $p = 2.99/3.20$, $p_{\text{tail}} = 2.53/2.64$.
- D7** Adjudication: diffusion falsified ($p_{\text{tail}} \leq 1.87$); single k^4 operator profile-falsified (convexity); composite sufficient; aliasing control $n_{\text{modes}} = n_z$; underflow guard re-derived.
- D8** Calibration: $D = 0.69 [0.29, 1.25]$, $\kappa = 0.011 [0.006, 0.020]$, $\sqrt{m} = 2.64 [2.25, 3.35]$, $k^* = 7.9 [5.1, 11.2]$; $\chi^2/\text{dof} = 3.6$; hand-fit rejected ($\Delta\chi^2 = 17.1$); sole residual mode-5 shelf ($z = -3.0$).
- D9** OOS test: field theory within 1% of empirical 3-factor on level/straddle/risk-reversal; diagonal fails two-sided; butterfly over-predicted $2.2\times$, independently localising the D8 residual.

B Reproducibility

All results derive from a public repository (github.com/MichaelAafton/diffusion-modes-vol-surface): a one-time WRDS pull script (raw data git-ignored under licence), the validated simulator and analytic forward model in `src/`, analysis scripts in `scripts/`, and a pytest suite that freezes the harness-recovery test, the increment-spectrum theory, and loader dtype guarantees. Randomness is seeded throughout; bootstrap and sweep settings are recorded in the scripts.

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